

Operating Systems – 21–22 Exam Answers (Organized)

Question 1

(a) Define Operating System. Explain computer system structure.

An operating system is system software that acts as an interface between the user and computer hardware and manages system resources such as CPU, memory, and input/output devices.

A computer system consists of hardware, operating system, system programs, and application programs. The operating system controls hardware and acts as an intermediary between users and hardware to ensure efficient and safe resource usage.

(b) Why OS is stored in firmware or disk?

Firmware storage provides fast startup and reliability for small, fixed systems such as embedded devices. Disk storage allows flexibility, frequent updates, and supports large modern operating systems.

(c) Interrupt and interrupt handling.

An interrupt is a signal generated by hardware or software requesting CPU attention. On an interrupt, the CPU saves its state, transfers control to an interrupt service routine, executes the routine, restores the state, and resumes execution.

(d) Client–Server vs Peer-to-Peer computing.

Client–Server computing is centralized, easier to manage, and depends on servers. Peer-to-peer computing is decentralized, has no single point of failure, and is harder to manage.

Question 2

(a) Services provided by an operating system.

Program execution, I/O operations, file system manipulation, communication, error detection, resource allocation, and protection and security.

(b) Two essential parts of a process.

A process consists of program code and a Process Control Block (PCB). A program is passive and stored on disk, whereas a process is active and has execution state and resources.

(c) Process Control Block (PCB).

PCB stores process ID, process state, program counter, CPU registers, scheduling information, memory management information, and I/O status.

(d) Direct Memory Access (DMA).

CPU initializes the DMA controller, which transfers data directly between device and memory. Limited interference may occur due to bus contention and cache coherency issues.

Question 3

(a) Storage device hierarchy.

Registers → Cache → Main Memory → Secondary Storage → Tertiary Storage. Closer storage is faster and more expensive; farther storage is slower and cheaper.

(b) Critical section problem.

The critical section problem ensures mutual exclusion, progress, and bounded waiting. It can be solved using mutexes, semaphores, monitors, or hardware instructions.

(c) Control flow of system calls.

A system call transfers control from user mode to kernel mode, executes the service, and returns to user mode. System calls are time-consuming due to mode switching and security checks.

(d) Advantages of multithreaded programming.

Improved responsiveness, better resource utilization, increased throughput, and efficient use of multi-core processors.

Question 4

(a) Deadlock conditions.

Mutual exclusion, hold and wait, no preemption, and circular wait must hold simultaneously for deadlock.

(b) CPU scheduling comparison.

For given processes, average waiting times are: FCFS = 28, SJF = 13, Round Robin = 23. SJF gives the minimum average waiting time.

(c) Scheduling criteria.

CPU utilization, throughput, turnaround time, waiting time, and response time.

Question 5

(a) Deadlock avoidance – arguments.

For: improved reliability and predictability. Against: increased overhead and turnaround time.

(b) Deadlock recovery.

Recovery can be done through process termination or resource preemption.

(c) Effective access time using TLB.

$$\text{EAT} = 0.8 \times 120 + 0.2 \times 220 = 140 \text{ ns.}$$

(d) Resource Allocation Graph.

All instances of required resources are allocated and processes form a circular wait; hence the system is deadlocked.

Question 6

(a) Banker's Algorithm – Need matrix and request.

Need = Max – Allocation. The request from P1 exceeds its need; hence it cannot be granted.

(b) Paging and address translation.

Paging divides memory into pages and frames. Logical address (page, offset) is mapped to physical address.

(c) Multithreaded word counting.

Two threads count words in two files concurrently; main thread combines results. Used in log analysis and document indexing.

Question 7

(a) Logical vs Physical address space.

Logical addresses are generated by CPU; physical addresses refer to actual memory locations.

(b) Fragmentation and segmentation.

Fragmentation wastes memory. Physical addresses are calculated as $\text{base} + \text{offset}$ if offset is valid.

(c) Address binding.

Address binding may occur at compile time, load time, or execution time using MMU.

Question 8

(a) FIFO page replacement.

For 3 frames, FIFO produces 13 page faults for the given reference string.

(b) Belady's Anomaly.

Increasing frames increases page faults in FIFO due to lack of stack property.

(c) Demand paging.

Pages are loaded only when required. Page faults trigger disk access and page table update.